INDEPENDENT TECHNICAL REVIEW — SUMMARY OF FINDINGS

Cure-cycle conformance evidence, Caldera CMS-1 composite shipsets

Prepared by Lyle Forsyth Associates for Halstead Aerospace Limited · 14 February 2025

1. Scope

Lyle Forsyth Associates was engaged on 24 January 2025 to conduct an independent review of the autoclave cure-cycle conformance records and first-article certification packs for shipsets S/N CMS1-0041 to CMS1-0058 supplied, or intended to be supplied, to Caldera Mobility Systems plc, following concerns raised internally regarding the accuracy of transcribed process-temperature data.

2. Findings

2.1 The review confirmed that, for nine of the eighteen shipsets examined, the peak cure temperatures recorded in the certification pack did not match the raw autoclave datalogger traces. In each case the recorded peak in the pack was within the qualified process window, whereas the raw trace showed brief excursions of up to 4°C below the minimum dwell temperature for part of the dwell period.

2.2 The transcription discrepancies appear to have arisen from manual entry from an interim summary report rather than from the validated datalogger export, and not from any deliberate falsification.

2.3 Coupon testing of retained witness panels from the affected runs returned interlaminar shear and short-beam strength results within specification, indicating that the affected parts are likely to be conforming notwithstanding the documentary discrepancy.

3. Recommendations

3.1 Re-issue the first-article certification packs for the nine affected shipsets using the validated datalogger export, with a concession note where an excursion is confirmed.

3.2 Notify Caldera Mobility Systems plc of the documentary discrepancy and the corrective action, and offer access to this review.

3.3 Implement an automated datalogger-to-pack export to remove the manual transcription step.

3.4 The review found no basis to doubt the good faith or competence of the individual who raised the original concern; on the contrary, the concern was well-founded as to the documentary discrepancy and was raised appropriately.

Dr Margaret Lyle, Director — Lyle Forsyth Associates